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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

IS3332-1 – DEEP LEARNING

(For IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **What is the purpose of a loss function in neural networks?**
A) It updates the weights after each epoch B) It determines the number of hidden layers
C) It measures the difference between actual and predicted values D) It prevents over fitting
2. **Which of the following loss functions is commonly used for regression problems?**
A) Cross-Entropy Loss B) Mean Squared Error (MSE)
C) Hinge Loss D) Binary Cross-Entropy
3. **What is the key benefit of hierarchical feature engineering in deep learning?**
A) It improves the performance of decision trees B) It eliminates the need for labelled data
C) It reduces the number of training parameters D) It allows models to learn simple and complex patterns in a structured manner
4. **Why autoencoders are considered unsupervised learning models?**
A) They do not require labelled data, for training B) They require labelled data for loss calculation
C) They perform supervised classification tasks D) They do not involve neural networks
5. **What is one of the main advantages of using stacked RBMs in deep learning?**
A) They eliminate the need for GPUs B) They enable pretraining of deep networks by learning hierarchical features
C) They do not require a large dataset for training D) They improve the speed of back propagation
6. **What is the main limitation of Hopfield Networks?**
A) They can only work with numerical data B) They cannot store multiple memory patterns
C) They have limited memory storage capacity D) They do not use activation functions
7. **What is the function of the forget gate in an LSTM?**
A) It decides which information to retain or discard from the cell state B) It controls the overall learning rate of the model
C) It updates the output of the network D) It replaces the activation function of LSTM
8. **What is an Echo-State Network (ESN)?**
A) A type of feed forward neural network B) A special form of RNN that uses a sparsely connected hidden layer with fixed random weights
C) A deep convolutional neural network D) A reinforcement learning model

9. What was a key improvement in VGG networks compared to AlexNet?
- A) It used deeper networks with smaller 3x3 convolutional filters
B) It introduced residual connections
C) It eliminated the need for pooling layers
D) It used recurrent layers instead of convolutional layers
10. What is the purpose of pooling layers in a CNN?
- A) To increase the number of parameters
B) To enhance the resolution of images
C) To replace convolutional layers
D) To reduce the spatial dimensions while retaining important features
11. Which of the following is NOT an application of RNNs?
- A) Sentiment analysis
B) Speech recognition
C) Image classification
D) Machine translation
12. What type of data is typically fed into the generator of a GAN?
- A) Raw images
B) Noise vectors
C) Labeled data
D) Preprocessed features
13. Which algorithm is designed to improve the policy directly in reinforcement learning without relying on value functions?
- A) Q-Learning
B) SARSA
C) Policy Gradient
D) Deep Q-Network
14. What does the term "exploitation" refer to in reinforcement learning?
- A) Choosing actions based on past experiences to maximize expected reward
B) Randomly selecting actions to gather more data
C) Ignoring rewards for short-term learning
D) Training on large labeled datasets
15. Which of the following is an example of a stateless reinforcement learning algorithm?
- A) Q-Learning
B) Monte Carlo methods
C) Multi-Armed Bandit
D) Deep Q-Networks
16. When should training typically stop when using early stopping?
- A) When training loss increases
B) When validation loss starts increasing
C) After a fixed number of epochs
D) When the model reaches 100% accuracy
17. Parameter sharing is commonly used in:
- A) Linear regression models
B) Convolutional Neural Networks (CNNs)
C) Decision Trees
D) K-Nearest Neighbors
18. Sparse representation in neural networks refers to:
- A) Representing data with dense matrices
B) Using many non-zero values in feature vectors
C) Increasing the dimensionality of data
D) Representing data with few active (non-zero) elements
19. Which of the following is an ensemble learning technique?
- A) Dropout
B) Bagging
C) L1 regularization
D) Batch normalization
20. Why is choosing an appropriate learning rate challenging in neural network optimization?
- A) Too small a learning rate causes slow convergence, while too large can cause divergence
B) A high learning rate always improves accuracy
C) A low learning rate leads to faster convergence
D) Learning rate has no effect on convergence

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	What is Deep Learning? List and explain the challenges motivating deep learning	8	L*2	1	1,2
b)	Define autoencoders and explain their basic principles. How do autoencoders perform dimensionality reduction, and in what ways are they similar to or different from Principal Component Analysis (PCA)?	8	L2	2	1,2

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|-------|--|---|----|---|-----|
| 2. a) | Explain the hierarchical engineering and pretrained model? How it is used in the Deep Learning? | 8 | L2 | 1 | 1,2 |
| b) | Write a short note on
i) Outlier detection using autoencoder
ii) Hopfield Networks | 8 | L2 | 2 | 1,2 |
| 3. a) | Define perceptron. With the help of a diagram, illustrate multilayer feed-forward network. | 8 | L2 | 1 | 1,2 |
| b) | With the help of a diagram, illustrate the Restricted Boltzmann machines (RBM)? Explain the applications of RBM. | 8 | L2 | 2 | 1,2 |

Unit – II

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|-------|--|---|----|---|-----|
| 4. a) | Explain the basic structure of a Convolutional Neural Network with a neat diagram. Discuss the process of training a CNN, including forward pass, backward pass, and optimization. | 8 | L3 | 3 | 1,2 |
| b) | What are Conditional GANs (CGANs)? How do they differ from standard GANs, and what advantages do they offer in controlled data generation? | 8 | L3 | 4 | 1,2 |
| 5. a) | What are Echo-State Networks (ESNs)? Discuss their unique characteristics and potential applications in processing time-series data. | 8 | L3 | 3 | 1,2 |
| b) | Explain the concept of stateless algorithms in reinforcement learning. How do they differ from stateful approaches, and in what situations are they applicable? | 8 | L3 | 4 | 1,2 |
| 6. a) | Discuss with the suitable example the limitations of neural networks? How it can be overcome? | 8 | L3 | 4 | 1,2 |
| b) | Discuss and Compare the architectures of AlexNet and VGG. | 8 | L3 | 3 | 1,2 |

Unit – III

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|-------|---|---|----|---|-----|
| 7. a) | Explain the working of dropout as a regularization technique in deep learning. Compare L1 and L2 regularization methods, highlighting their effects on model weights. | 8 | L2 | 5 | 1,2 |
| b) | Discuss how learning differs from optimization and why they are interconnected in deep learning. Explain the challenges of using gradient descent for very large datasets and how mini-batch gradient descent addresses these challenges. | 8 | L3 | 5 | 1,2 |
| 8. a) | Discuss the role of dataset augmentation as a regularization technique. Provide examples of augmentation methods and explain how they help prevent overfitting. | 8 | L3 | 5 | 1,2 |
| b) | Explain the Adam optimization algorithm. What are its key components, and how does it adapt learning rates for individual parameters? | 8 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
